SRN - PES2UG24CS410

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Software Engineering LAB01

Requirements Table:

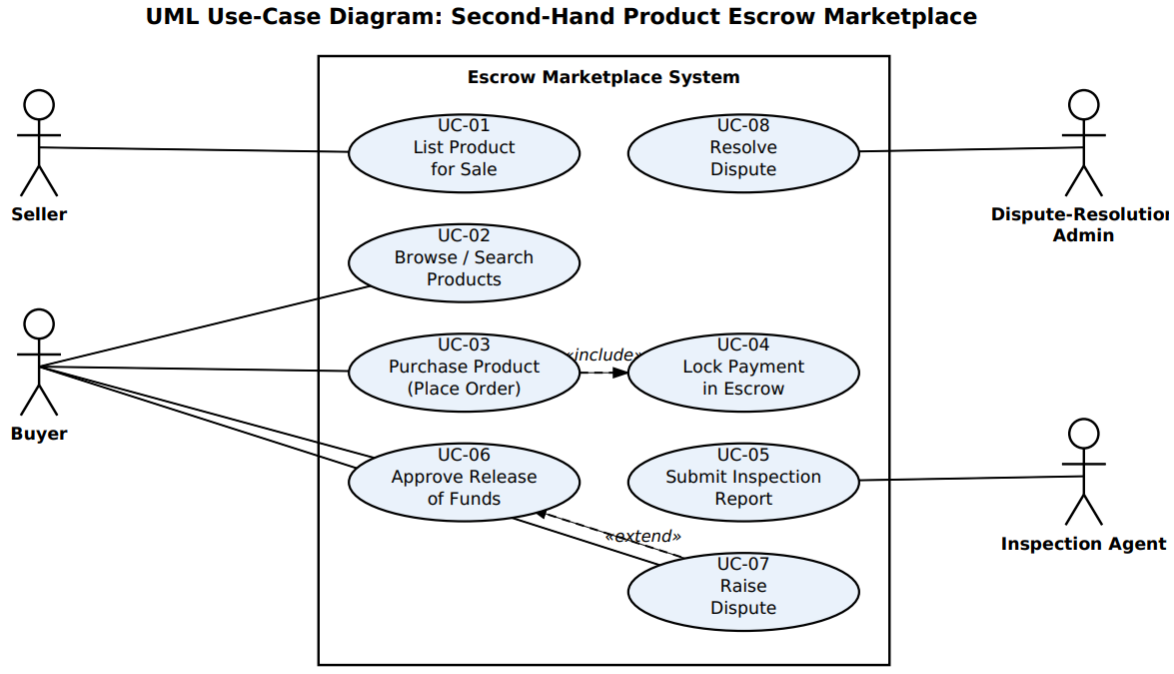
Problem Statement #39 – Second-Hand Product Escrow Marketplace

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall lock buyer payment in escrow and transition order state to 'Under Inspection' until an inspector submits physical verification sign-off.	High	Pass: Funds held securely and released only on dual verification. Fail: Funds released prematurely to seller.	Core trust mechanism of the escrow model; protects buyer funds until condition is independently verified.
FR-002	Functional	The system shall allow a seller to create a product listing with title, category, condition description, price, and at least one photo.	High	Pass: A listing submitted with all mandatory fields populated is published and searchable within 5 s. Fail: A listing missing a mandatory field is rejected with an inline error.	Complete, accurate listings let buyers make informed decisions and give inspectors a claimed-condition baseline to verify against.
FR-003	Functional	The system shall allow a buyer to search and filter listings by category, price range, and condition rating.	Medium	Pass: Applying a filter returns only listings matching all selected criteria, within 2 s. Fail: Results include any non-matching listing.	Enables buyers to efficiently locate relevant items among all active listings.
FR-004	Functional	The system shall allow an assigned Inspection Agent to submit a condition-verification report (checklist plus photos) against the listing's claimed condition.	High	Pass: A report with a completed checklist and at least one photo is timestamped and linked to the order. Fail: Submission with an incomplete checklist is blocked.	Provides the independent, auditable evidence required before escrow funds can be released.
FR-005	Functional	The system shall allow the buyer to approve release of escrow funds to the seller, or raise a dispute, within 48 hours of the inspection report being posted.	High	Pass: Buyer approval releases funds to the seller within 1 hour; a dispute freezes fund release and opens a dispute ticket. Fail: Funds release with no recorded buyer action and no fallback rule applied.	Gives the buyer final control over fund release while preventing the transaction from stalling indefinitely.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Nonfunctional – Performance & Security	Dispute resolution tickets must freeze escrow funds immediately and prevent account withdrawals during an active investigation.	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load.	Ensures a dispute cannot be circumvented by draining funds before resolution.
NFR-002	Nonfunctional – Security & Compliance	The system shall encrypt all payment and personally identifiable data in transit (TLS 1.2+) and at rest (AES-256), and shall store no raw card numbers (tokenized via a PCI-DSS-compliant gateway).	High	Pass: Security scan confirms TLS 1.2+ on all endpoints and no plaintext card data in the database. Fail: Any endpoint accepts a downgraded protocol, or plaintext card data is found in storage.	Protects sensitive financial data and satisfies baseline legal/compliance obligations for handling online payments.

Use-Case Diagram:

Problem Statement #39 – Second-Hand Product Escrow Marketplace



UseCase Flow Specification:

Problem Statement #39 – Second-Hand Product Escrow Marketplace

Preconditions:

Buyer is authenticated and has at least one verified payment method on file.

Selected product listing is Active and not already reserved by another buyer.

Seller account is in good standing (not suspended).

Postconditions (on success):

Payment amount is captured and held in the escrow account (not yet released to the seller).

Order status is set to 'Under Inspection'.

Listing status changes from Active to Reserved.

Main Success Scenario:

Buyer selects "Buy Now" on the product listing.

System displays an order summary: price, escrow terms, and estimated inspection timeline.

Buyer confirms the purchase and selects a payment method.

System validates the payment method and available funds.

System «include» invokes Lock Payment in Escrow (UC-04): authorizes and captures payment, transferring it into the secure escrow holding account.

System sets order status to 'Under Inspection' and notifies the assigned Inspection Agent.

System marks the listing as Reserved and notifies the seller that payment is secured.

Buyer receives an order confirmation with a reference/tracking ID.

Use case ends successfully; control passes to UC-05 Submit Inspection Report.

Alternate Flow – 4a. Payment Authorization Declined:

Payment gateway returns a decline response.

System displays a "payment failed" message; no funds are locked in escrow.

Buyer is prompted to retry with the same or a different payment method.

If payment fails after 2 consecutive attempts, the order is cancelled, the listing remains Active, and the buyer is notified. Use case ends in failure.

Related use cases: UC-04 Lock Payment in Escrow («include», mandatory step 5); UC-07 Raise Dispute («extend» of UC-06 Approve Release of Funds, downstream of this flow).

Use Case ID	UC-03
Use Case Name	Purchase Product (Place Order) – includes UC-04 «include» Lock Payment in Escrow
Primary Actor	Buyer
Secondary Actors	Payment Gateway (external system), Seller (notified), Inspection Agent (notified)
Trigger	Buyer selects “Buy Now” on an active product listing.